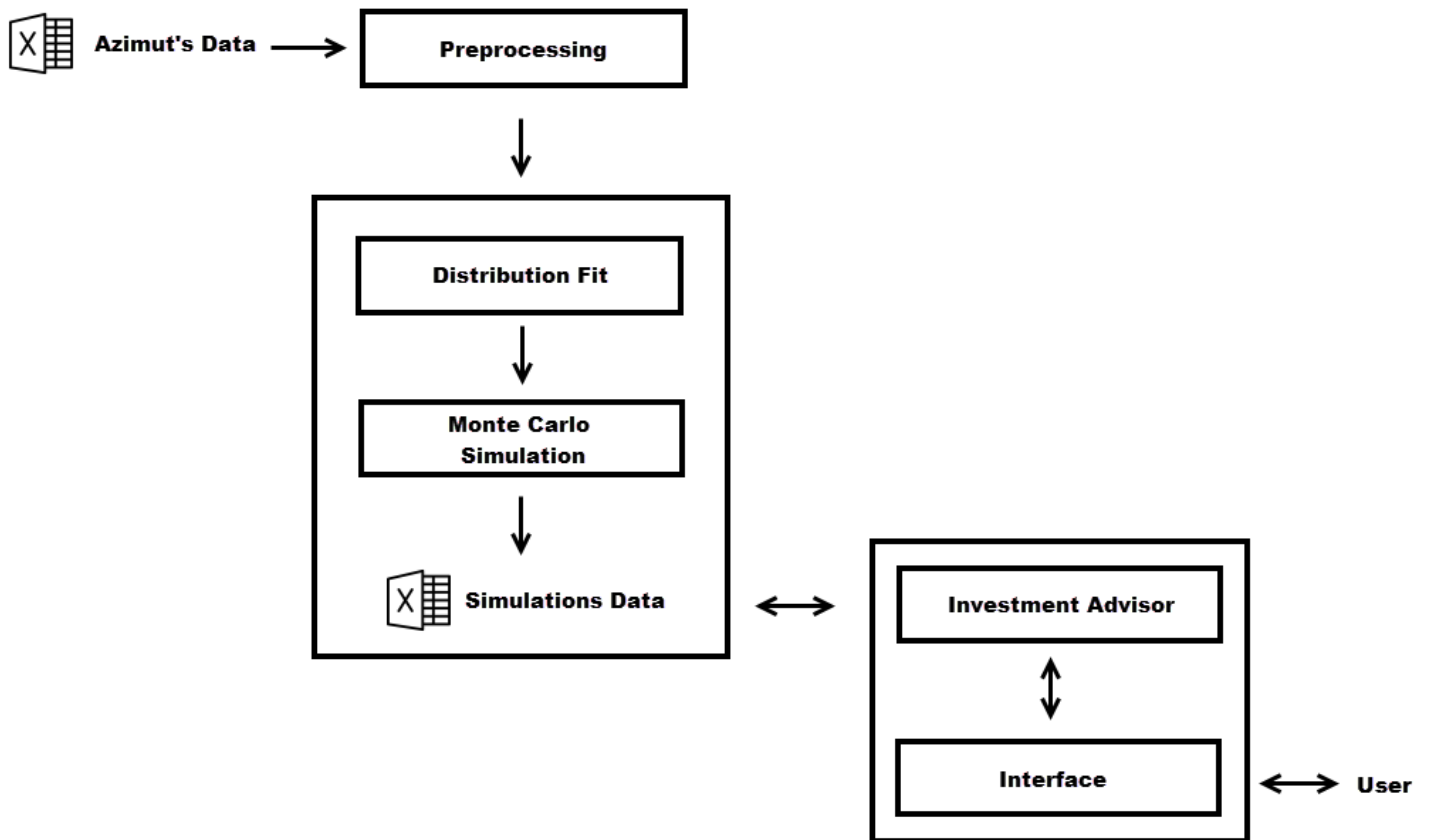


Robotic Financial Advisor Simulation

This document briefly explains the algorithm and rationale behind the simulation results for the Robotic Financial Advisor which you can test via the following link: <https://optofolio-rfa.herokuapp.com>

Overview:

As illustrated in the figure below, there are mainly three parts that contribute to the solution.



Part 1: Preprocessing

In this part we prepare the historical data required by the subsequent modules. We also do some analysis to study it but it's beyond the scope of this document.

For the inflation rates, we use the Month-on-month inflation data provided by Azimut and for the assets, we currently use Azimut's latest data provided for the [ABC Equity Fund](#) and the [EBE Money Market Fund](#).

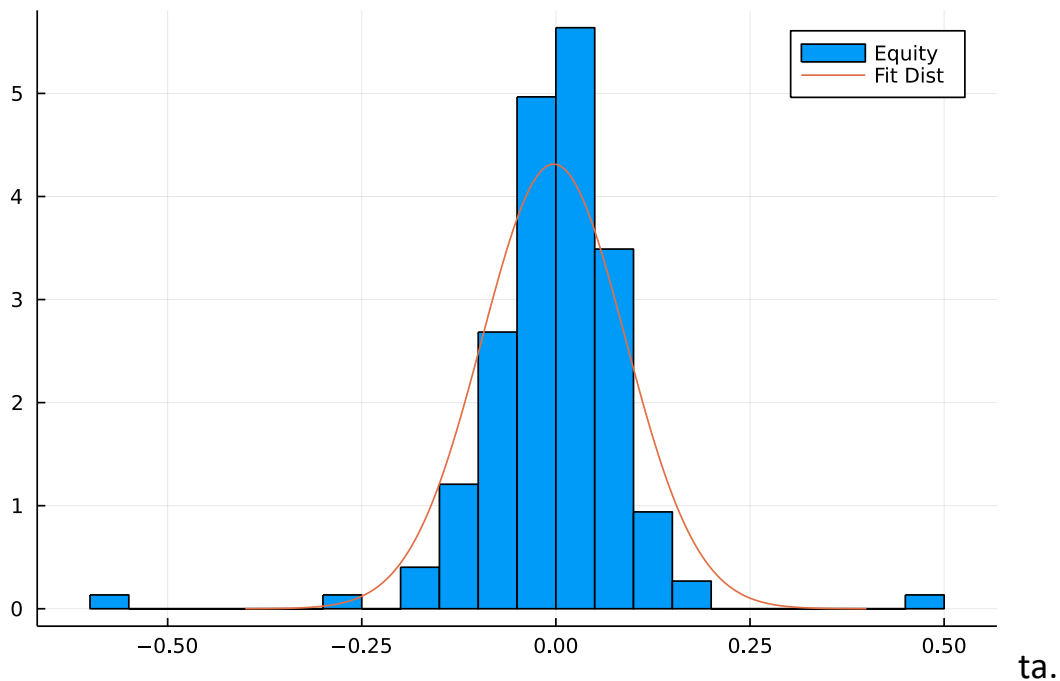
STEP 1: Since we're dealing with the data on a monthly basis, we take the latest data point for each month as illustrated in the figure below.

	category	Nav	MonthDate		category	Nav	MonthDate
1	"Sun Aug 10 2008"	100.0	► (2008, 8)	1	"Thu Aug 28 2008"	101.581	► (2008, 8)
2	"Thu Aug 21 2008"	100.03	► (2008, 8)	2	"Thu Sep 25 2008"	89.3807	► (2008, 9)
3	"Thu Aug 28 2008"	101.581	► (2008, 8)	3	"Thu Oct 30 2008"	66.5579	► (2008, 10)
4	"Thu Sep 04 2008"	102.281	► (2008, 9)	4	"Thu Nov 27 2008"	65.2415	► (2008, 11)
5	"Thu Sep 11 2008"	100.787	► (2008, 9)	5	"Thu Dec 25 2008"	68.715	► (2008, 12)
6	"Thu Sep 18 2008"	90.2727	► (2008, 9)	6	"Thu Jan 29 2009"	63.9637	► (2009, 1)
7	"Thu Sep 25 2008"	89.3807	► (2008, 9)	7	"Thu Feb 26 2009"	58.3452	► (2009, 2)
8	"Wed Oct 01 2008"	90.4212	► (2008, 10)	8	"Thu Mar 26 2009"	64.7888	► (2009, 3)
9	"Thu Oct 09 2008"	80.7776	► (2008, 10)	9	"Thu Apr 30 2009"	75.3461	► (2009, 4)
10	"Thu Oct 16 2008"	79.486	► (2008, 10)	10	"Thu May 28 2009"	77.2305	► (2009, 5)
11	"Thu Oct 23 2008"	69.3377	► (2008, 10)	11	"Thu Jun 25 2009"	77.2783	► (2009, 6)
12	"Thu Oct 30 2008"	66.5579	► (2008, 10)	12	"Thu Jul 30 2009"	85.1161	► (2009, 7)
13	"Thu Nov 06 2008"	72.4919	► (2008, 11)	13	"Thu Aug 27 2009"	92.6606	► (2009, 8)
14	"Thu Nov 13 2008"	71.5791	► (2008, 11)	14	"Thu Sep 24 2009"	96.6928	► (2009, 9)
15	"Thu Nov 20 2008"	63.2305	► (2008, 11)	15	"Thu Oct 29 2009"	93.7143	► (2009, 10)
	⋮				⋮		

STEP 2: We convert historical NAVs into nominal returns.

Part 2: Monte Carlo Simulation

STEP 1: We find a distribution that is a best fit to each asset's nominal returns. The figures below illustrate this for the equity fund data fit with a normal distribution with the following properties: ($\mu=0.006243724543093228$, $\sigma=0.09185267212882543$).



STEP 2: For 100 years that is 1200 months, we generate 10,000 scenarios as follows:

1. Sample from the fit distribution to the nominal fixed returns.
2. Sample from the fit distribution to the nominal variable returns.

STEP 3: We reduce the resulting matrices to a yearly format that is every consecutive 12 months are averaged.

Part 3: Interface

In this part, the main component is the **Investment Advisor** that translates the user's inputs into queries that run on the data generated in part 2. This separation is important for performance and consistency.

Most mappings between the user's input and the simulation input are obvious except for the goal importance and risk tolerance and we believe that the ambiguity around them can be a source of confusion.

Goal importance contributes to the criteria for picking the best portfolio and to illustrate this, let's consider the following results for several portfolios.

	Variable %	Worst (15 th Percentile)	Essential (30 th Percentile)	Med. (50 th Percentile)	Best (85 th Percentile)
1	90%	3,507,664	4,167,333	4,965,159	6,923,500
2	70%	3,760,153	4,308,673	4,917,612	6,370,171
3	10%	4,472,721	4,562,822	4,652,802	4,844,674

If the user chose the goal importance to be essential then the best allocation will be the one that yields the maximum portfolio value across 70% of the scenarios that is the 30th percentile and not the median.

Risk tolerance contributes to the number of portfolios the algorithm can test. Previously it directly mapped to the **maximum variable asset allocation** as follows:

Very Low	0%
Low	20%
Moderate	50%
High	80%
Very High	100%

This means that if the user had chosen their risk tolerance to be **Very High**, the algorithm would've tested all possible portfolios.

In the refinement, we modified it so that only 10% deviation of the user chosen risk tolerance mapped variable percentage is allowed. That is if you choose the Moderate risk option, the algorithm will consider a variable asset allocation of 60% +/- 10%.

Testing on the following scenario:

- **Goal:** Car
- **Duration:** 5 years
- **Importance:** Essential
- **Initial Investment:** 50,000 EGP
- **Monthly Investment:** 1,000 EGP
- **Annual Monthly Investment Increase:** 0%
- **Risk Tolerance:** High

The table below shows the portfolio values for several asset allocations:

	Variable %	Worst (15 th Percentile)	Essential (30 th Percentile)	Med. (50 th Percentile)	Best (85 th Percentile)
1	100%	73,336	90,818	118,943	211,032
2	90%	79,691	96,934	123,2456	206,584
3	80%	86,400	103,045	127,510	200,632
4	70%	93,481	109,231	131,051	194,936
5	0%	150,039	150,933	151,853	153,714

Because the **risk tolerance is high**, the algorithm will consider a variable asset allocation of **80% +/- 10%** that is 70%, 80% and 90%.

Because the **goal importance is essential**, the value chosen will be that of the **highest 30th percentile** that is a variable asset allocation of 70%.